WERGONIC · ASSESSMENT SCORING

RAMP & MEC — things that look wrong

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Below is a list of things in the RAMP and MEC scoring that look wrong to us. Some are almost certainly bugs. Others we are not sure about — we may be the ones who are wrong.

> **What we want from you:** go and look at the code, and tell us what it actually does. Where two places disagree, tell us which is which and where the difference comes from. A short answer per point is enough.
>
> **What we do not want:** do not go and decide what the ergonomically correct number is. That is not your call and you should not spend time on it — if a point turns out to depend on that, just say so and stop there. We will get the answer.

Items marked FIX are safe to change while you are in there. Items marked CHECK are report-back only — do not change behaviour.

NUMBERS THAT DISAGREE WITH EACH OTHER

1 Head bending backwards uses two different thresholds CHECK

The point where the head counts as leaning backwards is −10° in the risk verdict and in the Word report, but −5° on the phone. So the same worker gets a different answer depending on which screen you look at.

Trunk backwards is also −10°. That one is deliberate — leave it alone, and do not merge the two into one setting.

There is also a test in the repo asserting that −10 is the correct value for the head.

> **Tell us:** every place the head threshold is written, which output each one feeds, and when that test was added and by who.

2 One score is built differently from the other five CHECK

Head-backwards has 5 score steps. The other five posture scores have 7. It is also the only one that ever produces a 1.5 or a 6. There is a comment in that code saying it was rewritten at some point to match the official table.

> **Tell us:** confirm it really is the only one, and find the commit that changed it — who, when, and what the message says. Change nothing.

3 The same score gets two different colours CHECK

There are two separate pieces of code deciding the colour of a score and they do not agree. A 1.5 comes out yellow on the compare-assessments table and green on the single results page — same assessment, same number, two screens, two colours. Separately, a score of 1 is green everywhere except on question 1.8, where it is yellow.

> **Tell us:** where both colour rules live and what each one does, exactly. Do not pick a winner — we will tell you which one stays, then you apply it everywhere.

4 Hand distance ignores which task was selected CHECK

A RAMP for one task should only look at that task's time. Five of the six posture measures do. Hand distance uses the whole recording instead, so a result labelled as one task is mixed with the rest of the day.

Careful, this is bigger than it looks: that number is worked out once when the recording is uploaded and stored as a session total, so it probably cannot just be filtered afterwards.

> **Tell us:** confirm it, and give us a rough size for the fix before you start anything.

THINGS TO FIX

5 RAMP ignores the arm lengths the assessor typed in FIX

There is no sensor on the hand, so hand position is estimated from the arm angles and the length of the arm. The assessor can enter the worker's arm and forearm length when running an assessment. MEC uses what they entered. RAMP ignores it and always uses 32 cm and 27 cm, hardcoded — so the two assessments can disagree about the same worker.

> **Do:** make RAMP use the entered lengths, same as MEC. And label the number as an estimate wherever it is shown.

6 Trunk and head percentiles never reach the Word report FIX

They are calculated and stored correctly, then dropped at the last step because the report table only knows how to print arm rows.

> **Do:** add forward and backward bending rows for trunk and head. Leave side bending out.

7 A perfect result looks the same as a missing sensor FIX

If a worker never bends past the threshold, questions 1.1 and 1.2 show nothing at all — no score, no colour. That looks exactly like a question where the sensor was missing. Questions 1.3 to 1.6 show a green 0 in the same situation, which is what we want.

> **Do:** make 1.1 and 1.2 show the green 0 too, and check it gets counted in the green total at the top of the page.

8 Question 1.5 shows the wrong question text FIX

The heading is about upper arm posture, hand in or above shoulder height. The question printed underneath asks about bending the upper body forwards or to the side — that is question 1.3's text, copied word for word. So the ergonomist is asked about the back while the score underneath is about the arm. The help text is correct, only the question is wrong.

> **Do:** fix it in all five languages.

9 Every RAMP item's sub-rows restart at 1.1 FIX

"1.1" appears under item 1.1, again under 1.2, again under 1.3, and so on. They are not sub-questions, they are what we measured: forward bending, side bending, backward bending.

> **Do:** name them instead of numbering them.

SMALLER THINGS TO CONFIRM

10 MEC question 11 — three names, one calculation CHECK

The screen calls it "Neck bending". The code works it out as the head angle minus the body angle. The backend's own internal name for it suggests it should be the head's angle on its own.

In the same block: the head and trunk readings are paired by their position in the list rather than by time. If either sensor drops a reading, everything after it compares the head at one moment against the trunk at another, and the leftover readings are silently discarded. Everywhere else in the system pairs by time.

> **Tell us:** what the calculation actually is today, and why the earlier attempt at fixing the pairing was reverted.

11 MEC questions 12.1 and 12.2 look dead CHECK

They are listed in the backend but never calculated, and as far as we can tell nothing on the web side reads them, so no customer has ever seen a blank where they should be.

> **Tell us:** confirm nothing anywhere displays them. Then we decide whether to delete them or build them.

12 The AI section can vanish from the report without a word CHECK

If the AI call fails, the report is still generated, just without that section.

> **Tell us:** confirm that is what happens, and whether the user gets any indication at all. No change.

One short answer per point is enough. If a point turns out to hang on what the correct ergonomic value should be, say so and leave it — that part is on us, not you.